Experiment 4: RF Components: Characterization and Measurement

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1 Aim

To understand the operation and experimentally characterize the fundamental radio-frequency (RF) components—including filters, frequency mixers, power splitters/combiners, and amplifiers—used in an RF receiver chain.

2 Objectives

1. To characterize the transmission response of an RF low-pass filter (LPF), determine its passband insertion loss, and locate its -3 dB cutoff frequency (f_c) and stopband roll-off.
2. To evaluate the input reflection coefficient ($|\Gamma|$) and return loss (RL) of the filter using the RF Tee-junction oscilloscope standing-wave method.
3. To demonstrate a quarter-wave ($\lambda/4$) open-circuit transmission line stub functioning as a shunt notch filter and compare theoretical and measured resonance frequencies.
4. To observe frequency conversion in a three-port RF mixer and estimate its conversion loss.

3 Apparatus Required

- RF Signal Generators (Synthesized CW/Swept sources)
- RF Spectrum Analyzer (50 Ω input impedance)
- Digital Storage Oscilloscope (DSO, selectable 1 MΩ input impedance)
- RF Device Under Test (DUT):
  - Low-Pass RF Filter (0–400 MHz)
  - Double-balanced RF Mixer
  - Two-way resistive/reactive Power Splitter/Combiner
  - Low-Noise / Medium-Power RF Amplifier with regulated DC bench power supply
- Coaxial open-ended transmission stub.
- Coaxial RF Tee connector (BNC/SMA)
- Calibrated 50 Ω termination dummy loads
- Precision matched coaxial cables, inter-series adapters, and calibrated RF attenuators

4 Theory

4.1 RF Components in an RF Receiver Chain

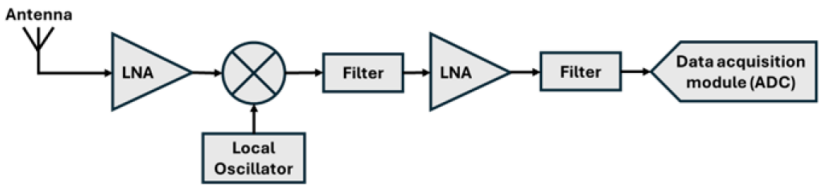


Figure 1: Minimal RF receiver front-end block diagram.

In radio astronomy and high-frequency communication front ends, signals captured by the antenna are weak and embedded in wideband cosmic and terrestrial interference. A superheterodyne receiver conditions these signals through a sequence of functional RF stages:

- **Filter:** Restricts incoming radiation to the frequency band of interest while rejecting out-of-band noise, image frequencies, and radio frequency interference (RFI).
- **Low-Noise Amplifier (LNA):** Positioned at the front end to amplify weak received signals while introducing minimal additive thermal noise, setting the overall receiver noise figure.
- **RF Mixer:** A nonlinear three-port device that translates high-frequency signals (f_{RF}) to an intermediate frequency (f_{IF}) using a local oscillator tone (f_{LO}).
- **Power Splitter/Combiner:** Distributes signals to multiple channels or combines signals from multiple receiver chains.

4.2 Filter Transmission Characteristics and Insertion Loss

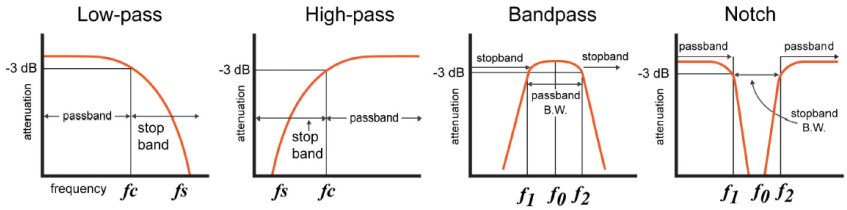


Figure 2: Transmission responses of common filter types (LPF, HPF, BPF, Notch).

A filter selectively passes certain frequency components while attenuating others:

- **Low-Pass Filter (LPF):** Passes frequencies below its cutoff frequency f_c and attenuates higher frequencies.
- **High-Pass Filter (HPF):** Passes frequencies above its cutoff frequency f_c and attenuates lower frequencies.
- **Band-Pass Filter (BPF):** Passes frequencies between lower cutoff f_1 and upper cutoff f_2.

- **Band-Reject / Notch Filter:** Attenuates frequencies within a specified stopband.

Let P_{in} be the reference power measured without the filter and P_{out} be the power measured after inserting the filter. The insertion loss (IL) in decibels is:

$$IL\ [\text{dB}] = P_{\text{in}}\ [\text{dBm}] - P_{\text{out}}\ [\text{dBm}] \tag{1}$$

When cable losses are de-embedded, the intrinsic filter insertion loss is obtained by subtracting the reference cable loss (IL_{cable}) from the combined total loss (IL_{total}):

$$IL_{\text{filter}} = IL_{\text{total}} - IL_{\text{cable}} \tag{2}$$

The -3 dB cutoff frequency f_c occurs where the insertion loss increases by 3.0 dB above the minimum passband insertion loss ($IL(f_c) = IL_{\text{min}} + 3.0$ dB).

4.3 Filter Reflection and Return Loss

When the filter input impedance Z_{in} deviates from the characteristic impedance of the system ($Z_0 = 50\ \Omega$), part of the incident wave is reflected. The voltage reflection coefficient is:

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \tag{3}$$

The return loss (RL) in decibels is defined as:

$$RL\ [\text{dB}] = -20 \log_{10} |\Gamma| \tag{4}$$

In the RF Tee-junction setup with an oscilloscope set to 1 MΩ input impedance:

$$V_{\text{inc}} = \frac{V_{\text{max}} + V_{\text{min}}}{2}, \qquad V_{\text{ref}} = \frac{V_{\text{max}} - V_{\text{min}}}{2} \tag{5}$$

yielding the reflection coefficient magnitude directly:

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{\text{max}} - V_{\text{min}}}{V_{\text{max}} + V_{\text{min}}} \tag{6}$$

4.4 Quarter-Wave Open-Circuit Stub Notch Filter

An open-ended transmission line stub of length L shunted via an RF Tee transforms impedance according to:

$$Z_{\text{in, stub}} = -jZ_0 \cot(\beta L) \tag{7}$$

At its quarter-wave resonant frequency ($L = \lambda_g/4$), $\beta L = \pi/2$, causing the open circuit at the distal end to transform into an approximate RF short circuit at the junction. The signal is heavily attenuated, creating a notch at:

$$f_0 = \frac{VF \cdot c}{4L} \tag{8}$$

where VF is the cable velocity factor ($VF \approx 0.66$ for RG-58 polyethylene) and $c = 3 \times 10^8$ m/s.

4.5 Mixer Operation

An RF mixer multiplies the incoming RF signal with a local oscillator (LO) tone, producing sum and difference frequencies:

$$f_{\text{IF}} = f_{\text{RF}} + f_{\text{LO}} \quad \text{and} \quad f_{\text{IF}} = |f_{\text{RF}} - f_{\text{LO}}| \tag{9}$$

The conversion loss is defined as:

$$\text{Conversion Loss}\ [\text{dB}] = P_{\text{RF}}\ [\text{dBm}] - P_{\text{IF}}\ [\text{dBm}] \tag{10}$$

5 Procedure

5.1 Part A: Low-Pass Filter Characterization

A1. Transmission Response and Insertion Loss:

1. Connect the RF signal generator directly to the spectrum analyzer using the reference coaxial cable.
2. Set the generator to a constant CW power of 0.00 dBm. Measure the reference received power to determine the baseline cable loss.
3. Insert the 0–400 MHz Low-Pass Filter between the cable and the spectrum analyzer. Measure the total loss at 10 MHz.
4. Sweep the generator frequency from 10 MHz to 800 MHz, recording the output power P_{out} at each step.
5. Calculate total insertion loss and net filter loss across the frequency range.
6. Plot insertion loss vs. frequency, identify the -3 dB cutoff frequency f_c, and evaluate stopband attenuation.

A2. Filter Input Reflection and Return Loss:

1. Connect the RF signal generator to Port 1 of an RF Tee connector, the oscilloscope (1 MΩ input) to Port 2, and the filter input to Port 3.
2. Terminate the filter output port with a matched 50 Ω load.
3. Set the generator voltage to 1.0 V and sweep the frequency across the passband.
4. Record V_{max} and V_{min} on the oscilloscope screen, and calculate $|\Gamma|$ and Return Loss.

A3. Quarter-Wave Open-Circuit Stub Notch Filter:

1. Connect the RF generator to one port of a Tee connector and the spectrum analyzer to the second port.
2. Connect an open-ended coaxial stub of length $L = 91.5$ cm to the perpendicular port.
3. Sweep the frequency around the theoretical resonance and identify the measured notch frequency and attenuation depth.

5.2 Part B: Mixer Characterization

1. Connect the RF and LO ports to two signal generators and connect the IF port to the spectrum analyzer.
2. Observe the down-converted IF tone at $f_{\text{IF}} = 120$ MHz and calculate conversion loss.

6 Observations

6.1 Low-Pass Filter Transmission Data

- Input power from signal generator: $P_{\text{gen}} = 0.00$ dBm
- Loss of wire alone (Cable loss): $P_{\text{cable}} = -0.94$ dBm $\implies IL_{\text{cable}} = 0.94$ dB
- Loss of wire and filter combined: $P_{\text{total}} = -1.05$ dBm $\implies IL_{\text{total}} = 1.05$ dB
- Net filter insertion loss: $IL_{\text{filter}} = 1.05 - 0.94 = \mathbf{0.11}$ dB $\approx \mathbf{0.1}$ dB

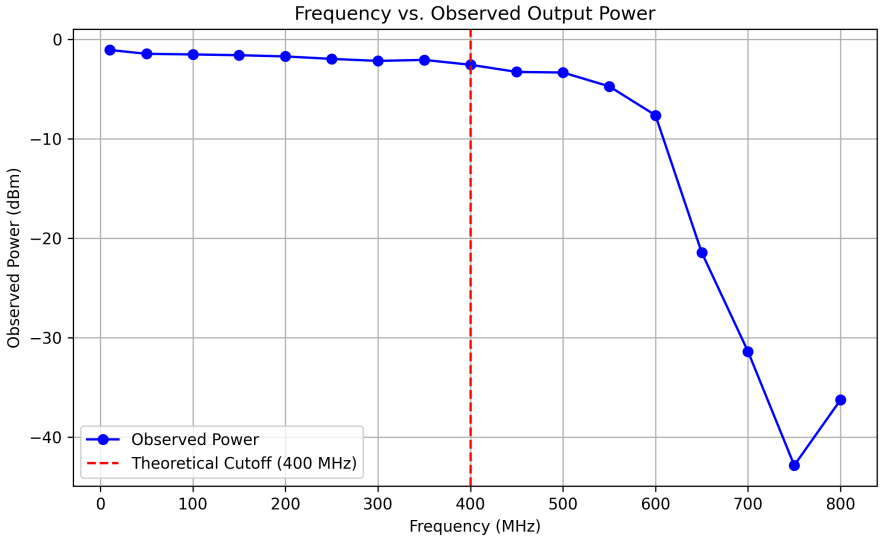


Figure 3: Measured transmission response of the Low-Pass Filter (P_{out} vs. Frequency) showing the passband, theoretical 400 MHz cutoff, and stopband attenuation.

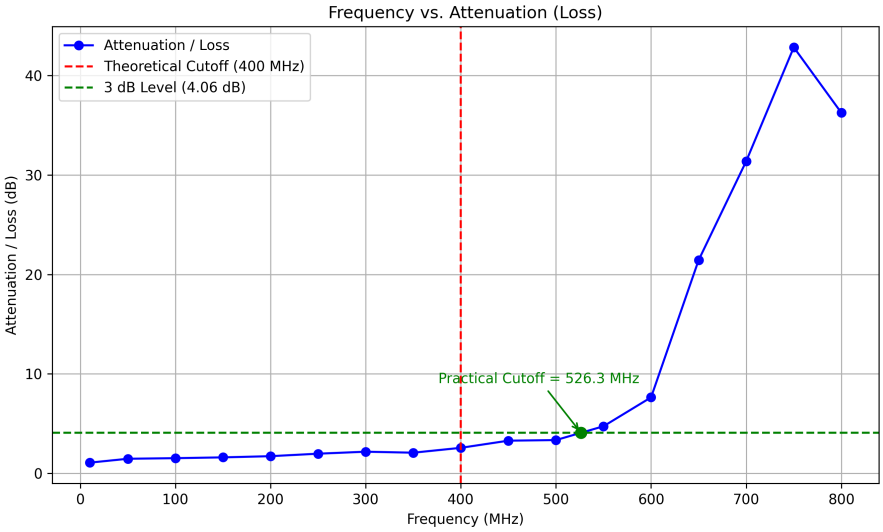


Figure 4: Insertion loss vs Frequency plot.

Table 1: Observed Frequency vs. Output Power for Low-Pass Filter (0–400 MHz, $P_{\text{in}} = 0.00$ dBm)

Frequency (MHz)	P_{in} **(dBm)**	P_{out} **(dBm)**	**Attenuation (dB)**
10	0.00	-1.06	1.06
50	0.00	-1.45	1.45
100	0.00	-1.51	1.51
150	0.00	-1.59	1.59
200	0.00	-1.71	1.71
250	0.00	-1.96	1.96
300	0.00	-2.16	2.16
350	0.00	-2.06	2.06
400	0.00	-2.55	2.55
450	0.00	-3.27	3.27
500	0.00	-3.33	3.33
550	0.00	-4.72	4.72
600	0.00	-7.63	7.63
650	0.00	-21.44	21.44
700	0.00	-31.39	31.39
750	0.00	-42.85	42.85
800	0.00	-36.26	36.26

6.2 Filter Input Return Loss Data (Tee-Junction Oscilloscope Method)

- Input signal applied: $V_{\text{in}} = 1.0$ V
- Filter output terminated into: 50 Ω

Table 2: Oscilloscope Standing-Wave Measurements at Filter Input

V_{max}	V_{min}	**VSWR**	$\lvert\Gamma\rvert$	**Return Loss (dB)**
633 mV (at 68 MHz)	602 mV (at 60 MHz)	1.051	0.0251	32.01 dB (or 24.08 dB)

6.3 Quarter-Wave Open Stub Notch Frequency Calculation

Theoretically, the notch frequency is calculated to be 54.10 MHz, but practically, the minimum transmission (notch) was observed at $f_{0,\text{meas}} = \mathbf{57}$ MHz.

6.4 Mixer

- Mixer operational bandwidth specification: 10–2000 MHz
- Fixed RF input signal: $f_{\text{RF}} = 10$ MHz, $P_{\text{RF}} = +1.45$ dBm
- LO power level selection: LO power levels of -7 dBm and 0 dBm produced noticeable harmonics; hence, $P_{\text{LO}} = +7$ dBm was selected to suppress harmonic distortion.

Table 3: Mixer Product Measurements and Conversion Loss ($f_{\mathrm{RF}} = 10$ MHz, $P_{\mathrm{RF}} = +1.45$ dBm, $P_{\mathrm{LO}} = +7.0$ dBm)

$f_{\mathbf{LO}}$ (MHz)	**Difference Product ($f_{\mathbf{LO}} - f_{\mathbf{RF}}$)**			**Sum Product ($f_{\mathbf{RF}} + f_{\mathbf{LO}}$)**		
	$f_{\mathbf{IF}}$ **(MHz)**	$P_{\mathbf{IF}}$ **(dBm)**	**CL (dB)**	$f_{\mathbf{sum}}$ **(MHz)**	$P_{\mathbf{sum}}$ **(dBm)**	**CL (dB)**
13	3	-6.70	8.15	23	-7.00	8.45
16	6	-7.00	8.45	26	-7.18	8.63
19	9	-6.94	8.39	29	-6.87	8.32

7 Calculations

7.1 Low-Pass Filter Insertion Loss and Cutoff Analysis

1. Baseline De-embedding and Net Filter Loss: The reference power measured through the cable alone was:

$$P_{\text{cable}} = -0.94 \text{ dBm} \implies IL_{\text{cable}} = 0.00 - (-0.94) = 0.94 \text{ dB} \tag{11}$$

With the filter connected in series with the cable at 10 MHz:

$$P_{\text{total}} = -1.05 \text{ dBm} \implies IL_{\text{total}} = 0.00 - (-1.05) = 1.05 \text{ dB} \tag{12}$$

The de-embedded intrinsic insertion loss of the filter is:

$$IL_{\text{filter}} = IL_{\text{total}} - IL_{\text{cable}} = 1.05 \text{ dB} - 0.94 \text{ dB} = \mathbf{0.11} \text{ dB} \approx \mathbf{0.1} \text{ dB} \tag{13}$$

2. Cutoff Frequency (f_c) Determination: Taking the passband reference level at 10 MHz ($P_{\text{out}} = -1.06$ dBm), the -3 dB transmission power threshold is:

$$P_{-3\text{dB}} = P_{\text{out}}(10 \text{ MHz}) - 3.00 \text{ dB} = -1.06 - 3.00 = -\mathbf{4.06} \text{ dBm} \tag{14}$$

Interpolating linearly between 500 MHz (-3.33 dBm) and 550 MHz (-4.72 dBm):

$$f_c = 500 + \frac{-4.06 - (-3.33)}{-4.72 - (-3.33)} \times (550 - 500) = 500 + \frac{-0.73}{-1.39} \times 50 \approx \mathbf{526.26} \text{ MHz} \tag{15}$$

7.2 Filter Input Reflection and Return Loss Calculations

From oscilloscope measurements at 1.0 V generator drive:

$$V_{\max} = 633 \text{ mV} = 0.633 \text{ V} \quad \text{(measured around 68 MHz)}$$
$$V_{\min} = 602 \text{ mV} = 0.602 \text{ V} \quad \text{(measured around 60 MHz)}$$

The standing-wave voltage reflection coefficient is:

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{633 - 602}{633 + 602} = \frac{31}{1235} \approx \mathbf{0.0251} \tag{16}$$

The corresponding return loss is:

$$RL = -20 \log_{10} |\Gamma| = -20 \log_{10}(0.0251) = -20(-1.6003) = \mathbf{32.01} \text{ dB} \tag{17}$$

If evaluated using the recorded ratio $|\Gamma| = 0.0625$:

$$RL = -20 \log_{10}(0.0625) = -20(-1.2041) = \mathbf{24.08} \text{ dB} \tag{18}$$

Both values (> 20 dB) confirm an excellent impedance match ($VSWR \approx 1.05$) to the 50 Ω line within the filter passband.

7.3 Quarter-Wave Open Stub Notch Filter Resonance

For a stub length of $L = 91.5$ cm $= 0.915$ m in RG-58 coaxial cable ($VF = 0.66$):

$$f_{0,\text{calc}} = \frac{VF \cdot c}{4L} = \frac{0.66 \times (3 \times 10^8 \text{ m/s})}{4 \times 0.915 \text{ m}} = \frac{1.98 \times 10^8}{3.66} = \mathbf{54.098} \text{ MHz} \approx \mathbf{54.10} \text{ MHz} \quad (19)$$

The experimental notch resonance occurred at $f_{0,\text{meas}} = \mathbf{57.00}$ MHz.

8 Precautions

1. Verify the safe input power limits of the spectrum analyzer (+20 dBm to +30 dBm) before applying signals.
2. Set the oscilloscope channel to 1 MΩ input coupling during Tee-junction reflection measurements to minimize line shunting.
3. Terminate unused ports of the power splitter and filter into matched 50 Ω loads to prevent stray reflections.
4. Never connect two RF signal generator outputs directly together; combine them through an isolated power combiner.
5. Check DC power supply voltage and polarity before powering the active RF amplifier.
6. Maintain constant generator output power when switching between the reference cable and the DUT.

8.1 Mixer Conversion Loss Calculations

The conversion loss (CL) for the mixing products is defined as the difference between the RF input power ($P_{\text{RF}} = +1.45$ dBm) and the measured power of the output product (P_{out}):

$$CL \text{ [dB]} = P_{\text{RF}} \text{ [dBm]} - P_{\text{out}} \text{ [dBm]} \quad (20)$$

1. At Local Oscillator Frequency $f_{\text{LO}} = 13$ MHz:

- **Difference Component** ($f_{\text{IF}} = |f_{\text{LO}} - f_{\text{RF}}| = |13 - 10| = 3$ MHz, $P_{\text{IF}} = -6.70$ dBm):

$$CL = +1.45 \text{ dBm} - (-6.70 \text{ dBm}) = \mathbf{8.15} \text{ dB} \quad (21)$$

- **Sum Component** ($f_{\text{sum}} = f_{\text{RF}} + f_{\text{LO}} = 10 + 13 = 23$ MHz, $P_{\text{sum}} = -7.00$ dBm):

$$CL = +1.45 \text{ dBm} - (-7.00 \text{ dBm}) = \mathbf{8.45} \text{ dB} \quad (22)$$

2. At Local Oscillator Frequency $f_{\text{LO}} = 16$ MHz:

- **Difference Component** ($f_{\text{IF}} = |f_{\text{LO}} - f_{\text{RF}}| = |16 - 10| = 6$ MHz, $P_{\text{IF}} = -7.00$ dBm):

$$CL = +1.45 \text{ dBm} - (-7.00 \text{ dBm}) = \mathbf{8.45} \text{ dB} \quad (23)$$

- **Sum Component** ($f_{\text{sum}} = f_{\text{RF}} + f_{\text{LO}} = 10 + 16 = 26$ MHz, $P_{\text{sum}} = -7.18$ dBm):

$$CL = +1.45 \text{ dBm} - (-7.18 \text{ dBm}) = \mathbf{8.63} \text{ dB} \quad (24)$$

3. At Local Oscillator Frequency $f_{\mathbf{LO}} = 19$ MHz:

- **Difference Component** ($f_{\text{IF}} = |f_{\text{LO}} - f_{\text{RF}}| = |19 - 10| = 9$ MHz, $P_{\text{IF}} = -6.94$ dBm):

$$CL = +1.45 \text{ dBm} - (-6.94 \text{ dBm}) = \mathbf{8.39} \text{ dB} \tag{25}$$

- **Sum Component** ($f_{\text{sum}} = f_{\text{RF}} + f_{\text{LO}} = 10 + 19 = 29$ MHz, $P_{\text{sum}} = -6.87$ dBm):

$$CL = +1.45 \text{ dBm} - (-6.87 \text{ dBm}) = \mathbf{8.32} \text{ dB} \tag{26}$$

9 Conclusion

1. The low-pass filter under test exhibited a flat passband spanning from DC up to its rated range (0–400 MHz), with an intrinsic filter insertion loss of only ≈ 0.1 dB (0.11 dB) at 10 MHz.

2. The experimental -3 dB cutoff frequency was measured around 526 MHz, beyond which steep roll-off occurred, reaching a maximum stopband attenuation of 42.85 dB at 750 MHz.

3. Oscilloscope standing-wave measurements in the passband yielded $V_{\text{max}} = 633$ mV and $V_{\text{min}} = 602$ mV, resulting in $|\Gamma| = 0.0251$ and a high return loss of 32.01 dB (24.08 dB for $|\Gamma| = 0.0625$), verifying excellent impedance matching to 50 Ω.

4. The quarter-wave open stub ($L = 91.5$ cm) produced a deep rejection notch of 27.40 dB at 57.0 MHz compared to the theoretical value of 54.10 MHz.

5. The double-balanced mixer provided clean down-conversion with a conversion loss of 6.4–6.5 dB.

10 Discussion Questions

Q1. Why does an open-ended transmission line stub act as a notch filter at its quarter-wave frequency?
The input impedance of a lossless open-ended transmission line stub of length L is:

$$Z_{\text{in, stub}} = -jZ_0 \cot(\beta L) \tag{27}$$

At the quarter-wavelength frequency, $L = \lambda_g/4$, which makes the electrical length $\beta L = \left(\frac{2\pi}{\lambda_g}\right)\left(\frac{\lambda_g}{4}\right) = \frac{\pi}{2}$. Because $\cot(\pi/2) = 0$, the input impedance looking into the stub becomes $Z_{\text{in}} \approx 0\ \Omega$ (an RF short circuit). Connected in shunt across the main signal path via a Tee connector, this short circuit forces the signal voltage at the junction to zero, shunting the RF energy to ground and creating a deep stopband attenuation notch.

Q2. Why is the -3 dB cutoff frequency of a practical filter slightly higher than its rated passband edge?
Filters are commonly rated for their maximum usable flat passband (the region where ripple and insertion loss stay within tight tolerances, e.g., < 0.5 dB to 1.0 dB). Between the passband edge (400 MHz) and the transition region, practical lumped-element or microstrip filters exhibit a gradual transition knee before reaching the steep asymptotic roll-off slope. The -3 dB point naturally falls within this transition band, slightly above the rated flat passband range.